EXPERIMENT 10

BER for BPSK over AWGN and Rayleigh Fading Channels with Correlated Channel Matrix

Date of Performance: _____ Date of Submission: _____ Marks: _____

1. AIM / OBJECTIVE

To compare the BER performance of BPSK modulation over AWGN and Rayleigh fading channels through theoretical analysis and Monte Carlo simulation, and to generate a spatially correlated channel matrix using a given transmit correlation matrix.

2. APPARATUS / SOFTWARE REQUIRED

- Personal Computer / Workstation
- MATLAB R2024a (or later version)
- Statistics and Machine Learning Toolbox (for mvnrnd function)

3. THEORY

AWGN Channel: The Additive White Gaussian Noise channel is the ideal baseline model. It adds white Gaussian noise to the signal with no fading or multipath effects. The theoretical BER for BPSK over AWGN is: Pb = Q(sqrt(2 Eb/N0)) = 0.5 erfc(sqrt(Eb/N0)).

Rayleigh Fading Channel: In environments with many scatterers and no dominant line-of-sight path (e.g., dense urban areas), the received signal envelope follows a Rayleigh distribution. This causes severe random amplitude fluctuations. The theoretical BER for BPSK over a Rayleigh channel is: Pb = 0.5 [1 - sqrt(Eb/N0 / (1 + Eb/N0))]. Rayleigh fading results in a much slower BER improvement with SNR (1/SNR behaviour) compared to AWGN (exponential improvement).

Monte Carlo Simulation: BER simulation involves generating random bit sequences, applying BPSK mapping, passing through the channel model (AWGN or Rayleigh), adding noise, and comparing decisions to the original bits. Averaging over a large number of bits gives an empirical BER estimate that is compared with the theoretical curve.

Correlated MIMO Channel Matrix: In multi-antenna (MIMO) systems, spatial correlation between antennas reduces the effective diversity gain. The transmit correlation matrix R_T models this dependency. A correlated channel matrix G is generated using the Cholesky factor of R_T: G = R_T^(1/2) H, where H contains i.i.d. complex Gaussian entries. The MATLAB function mvnrnd() is used to draw samples from the multivariate normal distribution defined by R_T. Higher correlation (values of R_T closer to 1) degrades diversity order and worsens BER performance.

4. CONCLUSION

The experiment clearly demonstrates the severe BER degradation caused by Rayleigh fading compared to the AWGN channel. Simulated BER curves closely match the theoretical results, validating the Monte Carlo approach. The AWGN channel achieves extremely low BER at high Eb/N0 due to its exponential improvement, while the Rayleigh channel shows only a slow 1/SNR improvement. The correlated channel matrix generation illustrates the impact of transmit antenna correlation in MIMO systems, highlighting the

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```
clc;
clear;
close all;
```

========== PARAMETERS ==========

```
N = 1e6;                        % Number of bits
Eb_N0_dB = -3:2:20;             % SNR range (dB)
Eb_N0 = 10.^(Eb_N0_dB/10);      % Linear scale
```

========== TRANSMITTER ==========

```
ip = rand(1,N) > 0.5;           % Random bits
s = 2*ip - 1;                   % BPSK (0→-1, 1→+1)
```

========== AWGN CHANNEL ==========

```
simBer_awgn = zeros(size(Eb_N0_dB));

for i = 1:length(Eb_N0_dB)
    n = (1/sqrt(2))*(randn(1,N) + 1j*randn(1,N)); % noise
    y = s + (1/sqrt(Eb_N0(i))) * n;

    ipHat = real(y) > 0;
    simBer_awgn(i) = sum(ip ~= ipHat)/N;
end

% Theoretical BER (AWGN)
theory_awgn = 0.5 * erfc(sqrt(Eb_N0));
```

========== RAYLEIGH CHANNEL ==========

```
simBer_rayleigh = zeros(size(Eb_N0_dB));

for i = 1:length(Eb_N0_dB)
    n = (1/sqrt(2))*(randn(1,N) + 1j*randn(1,N));
    h = (1/sqrt(2))*(randn(1,N) + 1j*randn(1,N)); % Rayleigh channel

    y = h .* s + (1/sqrt(Eb_N0(i))) * n;
```

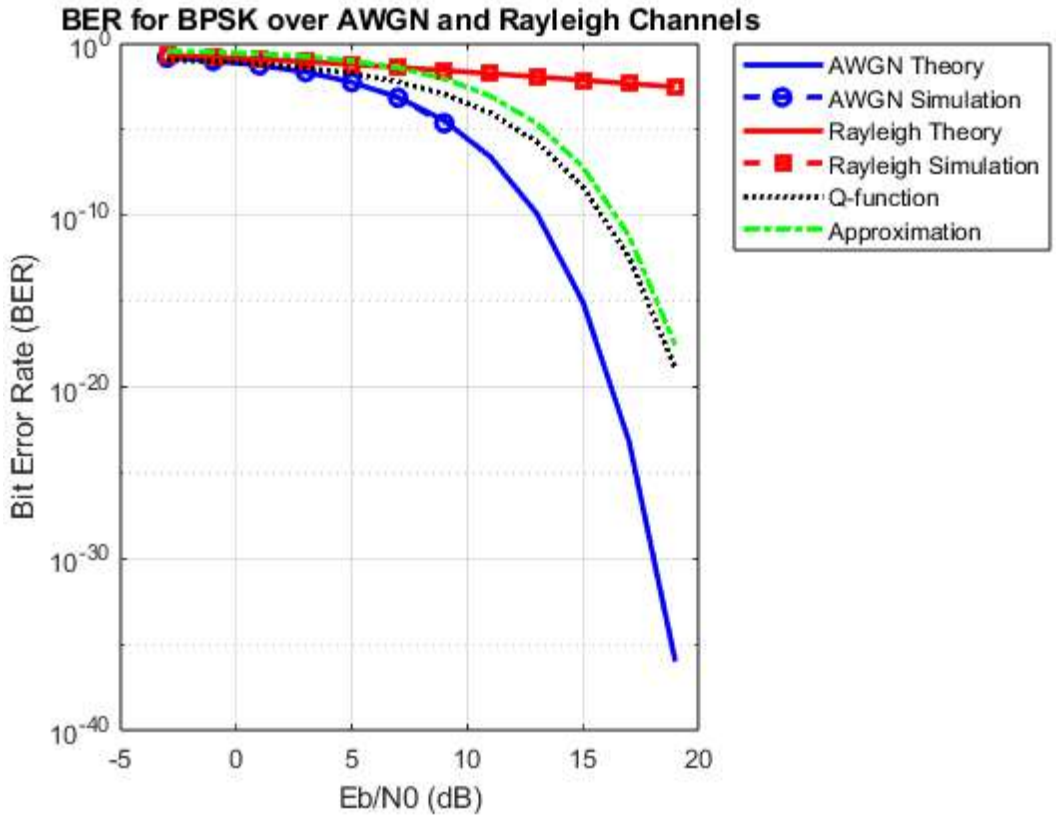
```
    yHat = y ./ h; % Equalization
    ipHat = real(yHat) > 0;

    simBer_rayleigh(i) = sum(ip ~= ipHat)/N;
end

% Theoretical BER (Rayleigh)
theory_rayleigh = 0.5 * (1 - sqrt(Eb_N0 ./ (1 + Eb_N0)));
```

========== EXTRA BER EXPRESSIONS ==========

```
BER_qfunc = 0.5 * qfunc(sqrt(Eb_N0));
BER_approx = 0.5 * exp(-Eb_N0/2);
```

========== PLOT ==========

```
figure;

semilogy(Eb_N0_dB, theory_awgn, 'b-', 'LineWidth', 2); hold on;
semilogy(Eb_N0_dB, simBer_awgn, 'bo--', 'LineWidth', 2);

semilogy(Eb_N0_dB, theory_rayleigh, 'r-', 'LineWidth', 2);
semilogy(Eb_N0_dB, simBer_rayleigh, 'rs--', 'LineWidth', 2);

semilogy(Eb_N0_dB, BER_qfunc, 'k:', 'LineWidth', 2);
semilogy(Eb_N0_dB, BER_approx, 'g-.', 'LineWidth', 2);

grid on;
xlabel('Eb/N0 (dB)');
ylabel('Bit Error Rate (BER)');
title('BER for BPSK over AWGN and Rayleigh Channels');

%%Legend moved to right side
legend('AWGN Theory','AWGN Simulation', ...
       'Rayleigh Theory','Rayleigh Simulation', ...
       'Q-function','Approximation', ...
       'Location','northeastoutside');
```



========== CORRELATED CHANNEL MATRIX ==========

```
R_T = [1 0.7; 0.7 1];

g1 = mvnrnd([0 0], R_T).';
g2 = mvnrnd([0 0], R_T).';

G = [g1 g2];

disp('Correlated Channel Matrix G:');
disp(G);
```

```
Correlated Channel Matrix G:
   -0.0191   -0.0077
   -1.8379    0.4144
```

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